

GNOSS FIELD, CA, USA

WMO: 720406

Lat: 38.150N

Lon: 122.550W

Elev: 1

StdP: 101.31

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00135

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.3	0.2	-7.2	2.0	14.5	-3.9	2.7	10.8	10.2	14.2	8.4	12.4	0.5	190	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	34.7	19.1	32.5	18.8	30.1	18.2	20.7	31.4	19.9	29.8	19.1	28.4	3.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.4	22.7	16.2	11.5	22.7	15.2	10.8	21.4	59.4	31.6	56.6	29.8	53.8	28.5	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.8	DB	-3.7	38.6	1.4	2.0	-4.7	40.0	-5.6	41.2	-6.4	42.3	-7.4	43.7	(4)
(5)			WB	-3.9	22.5	1.3	0.7	-4.9	23.0	-5.6	23.5	-6.3	23.9	-7.2	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	9.3	10.7	12.1	13.9	15.7	18.5	19.6	19.8	19.3	16.6	12.2	8.8	(6)	
	DBStd	4.73	2.43	2.68	2.55	2.82	2.63	2.88	2.26	2.20	2.69	2.62	2.81	2.95	(7)	
	HDD10.0	146	42	22	9	2	0	0	0	0	0	0	12	58	(8)	
	HDD18.3	1531	281	215	193	134	89	31	11	9	18	67	186	296	(9)	
	CDD10.0	1871	19	40	74	121	177	256	299	302	280	205	76	21	(10)	
	CDD18.3	215	0	0	0	3	9	36	52	54	48	13	0	0	(11)	
	CDH23.3	3702	1	6	14	113	243	679	783	742	782	331	9	1	(12)	
	CDH26.7	1531	0	0	1	31	91	306	319	307	369	107	0	0	(13)	
(14) Wind	WSAvg	2.3	1.9	2.4	2.4	2.8	2.9	2.6	2.5	2.2	2.2	1.9	1.7	1.9	(14)	
(15) Precipitation	PrecAvg	881	173	173	93	49	34	6	0	2	2	41	90	193	(15)	
	PrecMax	1315	524	531	332	166	104	23	1	27	6	131	248	441	(16)	
	PrecMin	448	15	9	0	5	0	0	0	0	0	0	5	18	(17)	
	PrecStd	252	128	118	94	41	33	8	0	6	2	38	71	132	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	23.8	25.1	30.0	32.6	37.0	36.0	37.1	37.2	32.6	24.2	18.8	(19)	
		MCWB	10.9	14.5	16.0	18.1	18.5	20.8	19.1	19.7	18.7	16.9	15.1	9.3	(20)	
	2%	DB	17.6	20.8	22.5	27.1	29.0	32.9	32.6	32.8	33.9	29.7	22.6	17.2	(21)	
		MCWB	11.5	14.2	14.9	16.6	17.4	19.3	19.6	19.2	18.3	15.9	14.3	11.9	(22)	
	5%	DB	15.9	18.0	20.0	23.0	25.9	30.1	30.1	29.9	31.2	27.3	20.2	15.2	(23)	
		MCWB	11.5	12.4	13.6	15.3	16.3	18.2	19.1	18.8	17.9	15.4	13.8	11.8	(24)	
	10%	DB	14.0	16.3	17.8	20.2	22.7	27.4	27.7	27.5	27.7	24.7	17.9	14.0	(25)	
		MCWB	11.3	12.4	12.7	14.1	15.2	17.6	18.6	18.5	17.6	15.1	12.7	11.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	16.1	17.3	19.2	20.2	22.0	21.5	21.2	20.7	19.4	17.4	15.8	(27)	
		MCDB	16.5	20.6	23.3	27.4	30.1	34.0	32.8	32.6	32.2	26.2	22.6	16.6	(28)	
	2%	WB	13.9	15.0	15.7	17.4	18.2	20.5	20.4	20.1	19.6	18.0	16.3	14.6	(29)	
		MCDB	15.5	18.6	20.2	25.1	27.7	31.1	30.7	30.3	30.0	25.0	19.9	15.5	(30)	
	5%	WB	12.8	14.0	14.7	16.0	16.9	19.2	19.5	19.4	18.7	17.2	15.2	13.5	(31)	
		MCDB	14.4	17.4	18.4	22.5	24.5	28.5	28.9	28.8	28.5	23.5	18.0	14.2	(32)	
	10%	WB	12.0	12.8	13.7	14.7	15.5	17.9	18.6	18.5	17.8	16.2	14.0	12.5	(33)	
		MCDB	13.5	15.4	16.9	19.3	22.0	25.9	27.6	27.2	26.2	22.1	16.9	13.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.9	11.2	12.4	13.4	16.1	17.1	16.8	17.9	15.9	12.6	10.0	(35)	
		MCDBR	13.6	15.0	15.7	18.2	20.1	22.3	21.3	21.8	24.0	20.5	15.8	11.7	(36)	
		MCWBR	8.4	8.8	8.9	9.6	9.9	9.9	9.6	9.5	9.8	9.1	8.5	7.4	(37)	
	5% WB	MCDBR	9.2	12.4	12.9	16.3	17.8	19.9	19.7	19.2	19.9	16.4	12.2	6.5	(38)	
		MCWBR	7.0	7.9	8.0	9.2	9.5	9.8	9.4	9.0	9.1	8.3	7.5	5.3	(39)	
(40) Clear-Sky Solar Irradiance	taub	0.320	0.324	0.334	0.354	0.359	0.355	0.361	0.341	0.341	0.341	0.325	0.318	0.321	(40)	
	taud	2.560	2.550	2.518	2.428	2.444	2.464	2.445	2.502	2.496	2.581	2.573	2.562		(41)	
	Ebn at Noon	866	909	930	925	920	920	912	926	909	896	861	839		(42)	
	Edh at Noon	75	86	97	112	113	111	112	103	98	82	73	71		(43)	
(44) All-Sky Solar Radiation	RadAvg	2.33	3.26	4.61	6.08	7.11	7.82	7.50	6.66	5.59	4.05	2.75	2.02		(44)	
	RadStd	0.35	0.43	0.45	0.50	0.45	0.35	0.26	0.26	0.21	0.28	0.19	0.32		(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	-1.21	N/A	N/A	N/A	N/A	-108	+135	N/A

Nomenclature: See separate page